

Exercise Set 3

Problem 1

A research project investigates the relationship between smoking habits and gender. The probability of smoking is described by the following equation:

$$P(\text{smoke} = 1) = \beta_0 + \beta_1 \text{female} + \beta_2 \text{smkban} + \beta_3 \text{age} + u. \quad (1)$$

The variables are defined as follows:

- smoke* = 1 if the person is a smoker, 0 otherwise;
- female* = 1 if the person is a female, 0 otherwise;
- smkban* = 1 if there is a smoking ban at the place of residence;
- age* = age in years.

u denotes an i.i.d. error term.

1.1 One option to estimate the probability $P(\text{smoke} = 1)$ is the linear probability model. What are the advantages and disadvantages of using a linear probability model?

Advantages:

- Easy estimation procedure.
- Interpretation as marginal effects.

Disadvantages:

- The predicted probabilities may be outside $[0,1]$, which contradicts the definition of a probability.
- The variance of the error term depends on the characteristics of the independent variables (heteroskedasticity).
- The error terms are not normally distributed, thus test statistics are invalid in small samples.

1.2 To achieve a better fit of the model to the data, one can estimate the same relationship using a logit model. Set up the likelihood function and the log-likelihood function for the logit model.

Likelihood function:

$$\begin{aligned} L &= \prod_{i=1}^N \Pr(\text{smoke}_i = 1)^{y_i} \Pr(\text{smoke}_i = 0)^{1-y_i} \\ &= \prod_{i=1}^N \left(\frac{\exp(\mathbf{x}'\beta)}{1 + \exp(\mathbf{x}'\beta)} \right)^{y_i} \left(\frac{1}{1 + \exp(\mathbf{x}'\beta)} \right)^{1-y_i} \end{aligned}$$

Log-likelihood function:

$$\begin{aligned}\ln L = \ln(L) &= \sum_{i=1}^N \{y_i \ln(\exp(\mathbf{x}'_i \boldsymbol{\beta})) - \ln(1 + \exp(\mathbf{x}'_i \boldsymbol{\beta}))\} \\ &= \sum_{i=1}^N \{y_i \mathbf{x}'_i \boldsymbol{\beta} - \ln(1 + \exp(\mathbf{x}'_i \boldsymbol{\beta}))\}\end{aligned}$$

1.3 Estimating the logit model with Stata yields the results reported in Table 1. Interpret the coefficient of *smkban*. Why would we expect a negative coefficient?

Table 1: regression results: Logit model

Logistic regression	Number of obs	=	10,000		
	LR chi2(3)	=	90.82		
	Prob > chi2	=	0.0000		
Log likelihood = -5491.7578	Pseudo R2	=	0.0082		

smoke	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]

female	-.0987092	.0473998	-2.08	0.037	-.1916112 -.0058072
smkban	-.3983811	.0475606	-8.38	0.000	-.4915982 -.305164
age	-.0059029	.0019526	-3.02	0.003	-.00973 -.0020759
_cons	-.6256074	.0843635	-7.42	0.000	-.7909569 -.4602579

- The sign of the estimated coefficient of *smkban* indicates that the log-odds for the event of smoking decrease, *ceteris paribus*, when a smoking ban exists.
- The *z* statistic (*z* = -8.38) and the very small *p* value show that the coefficient is statistically significantly different from zero.
- The negative effect for the variable *smkban* is expected because a smoking ban should lead to fewer smoking opportunities, making it harder to be a smoker.

1.4 Calculate the probability that a 30-year-old man smokes despite an existing smoking ban. Hint:

$$P(y_i = 1) = \frac{\exp(\mathbf{x}'_i \boldsymbol{\beta})}{1 + \exp(\mathbf{x}'_i \boldsymbol{\beta})}.$$

$$\begin{aligned}P(\text{smoke} = 1 | \text{female} = 0, \text{smkban} = 1, \text{age} = 30) &= \frac{\exp(-0.626 - 0.099 \cdot 0 - 0.398 \cdot 1 - 0.006 \cdot 30)}{1 + \exp(-0.626 - 0.099 \cdot 0 - 0.398 \cdot 1 - 0.006 \cdot 30)} \\ &= \frac{\exp(-1.204)}{1 + \exp(-1.204)} \\ &= \frac{0.300}{1.300} \\ &= 0.231\end{aligned}$$

1.5 Calculate and interpret the marginal effect of *age* for a 30-year-old man who smokes despite an existing smoking ban. *Hint:*

$$\frac{\partial P(y_i = 1)}{\partial x_{ik}} = \frac{\beta_k \exp(\mathbf{x}_i' \boldsymbol{\beta})}{(1 + \exp(\mathbf{x}_i' \boldsymbol{\beta}))^2}.$$

$$\begin{aligned} \frac{\partial P(y_i = 1)}{\partial age_i} &= \frac{\exp(-0.626 - 0.099 \cdot 0 - 0.398 \cdot 1 - 0.006 \cdot 30)}{(1 + \exp(-0.626 - 0.099 \cdot 0 - 0.398 \cdot 1 - 0.006 \cdot 30))^2} \cdot (-0.006) \\ &= \frac{\exp(-1.204)}{(1 + \exp(-1.204))^2} \cdot (-0.006) \\ &= \frac{0.300}{1.300^2} \cdot (-0.006) \\ &= -0.0011 \end{aligned}$$

An additional year of life c. p. reduces the probability of smoking by 1.1 percentage points for a 30-year-old male when a smoking ban is in place.

References

- Mroz, Thomas A. (1987). *The Sensitivity of an Empirical Model of Married Women's Hours of Work to Economic and Statistical Assumptions*. *Econometrica*, 55, 765-799.
- Wooldridge, Jeffrey M. (2013). *Introductory Econometrics. A modern Approach*. South-Western, Cengage Learning.